

Microcontroller

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Course ID: CSE - 4619

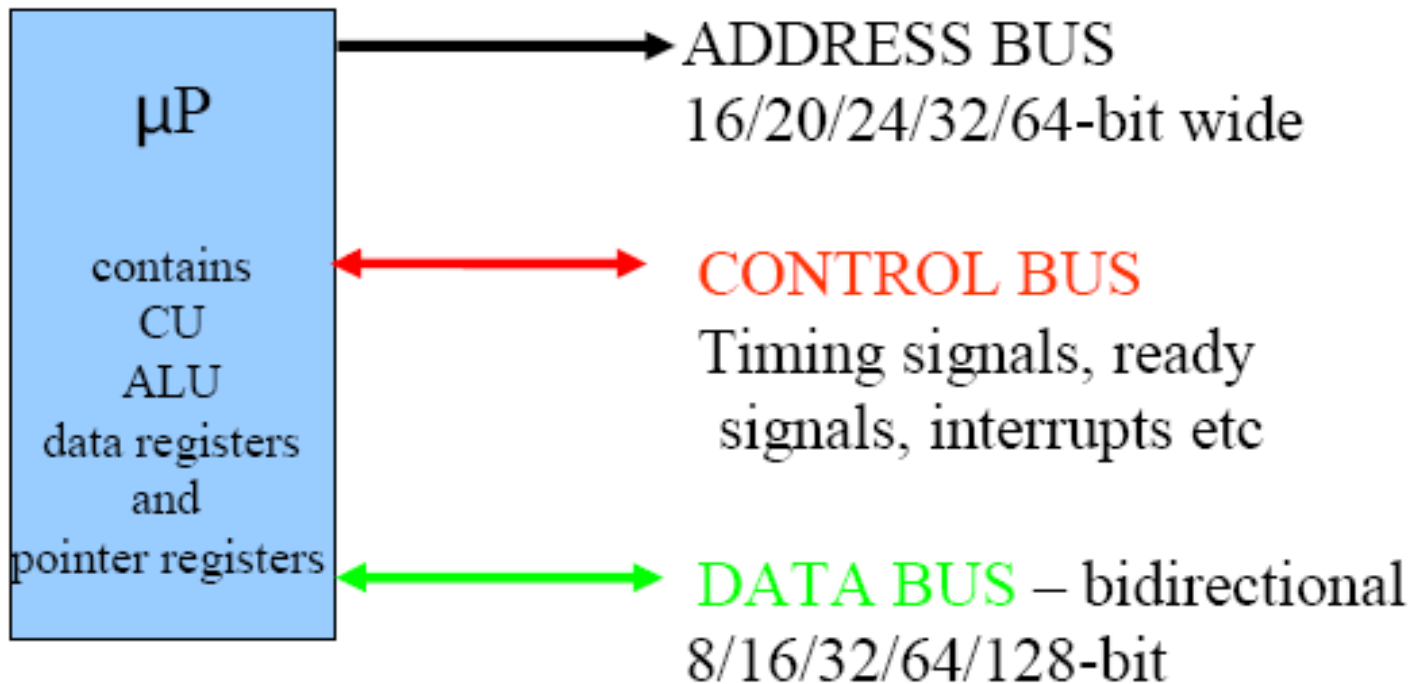
Course Title: Peripherals, Interfacing and Embedded Systems
Department of Computer Science and Engineering (CSE),
Islamic University of Technology (IUT), Gazipur.

Lecture References:

▶ **Web Materials:**

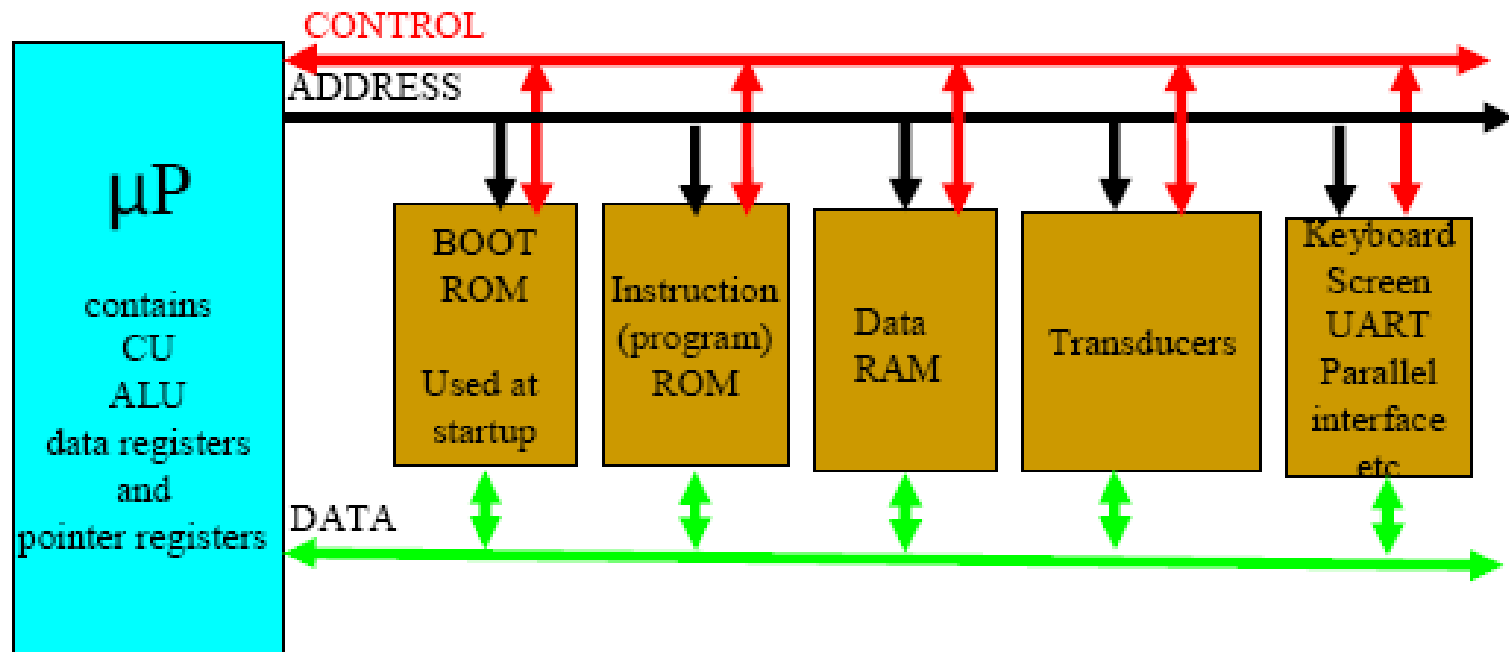
- ▶ http://en.wikipedia.org/wiki/Atmel_AVR
- ▶ www.atmel.com
- ▶ <http://www.atmel.com/Images/2466S.pdf>
- ▶ www.microchip.com
- ▶ www.alldatasheet.com
- ▶ www.avrfreaks.net
- ▶ www.arduino.cc

Microprocessor – Basic Concept



Microprocessor – Basic Concept

- ▶ Microprocessor by itself is completely useless.
 - ▶ MP must have external peripherals to interact with outside world.



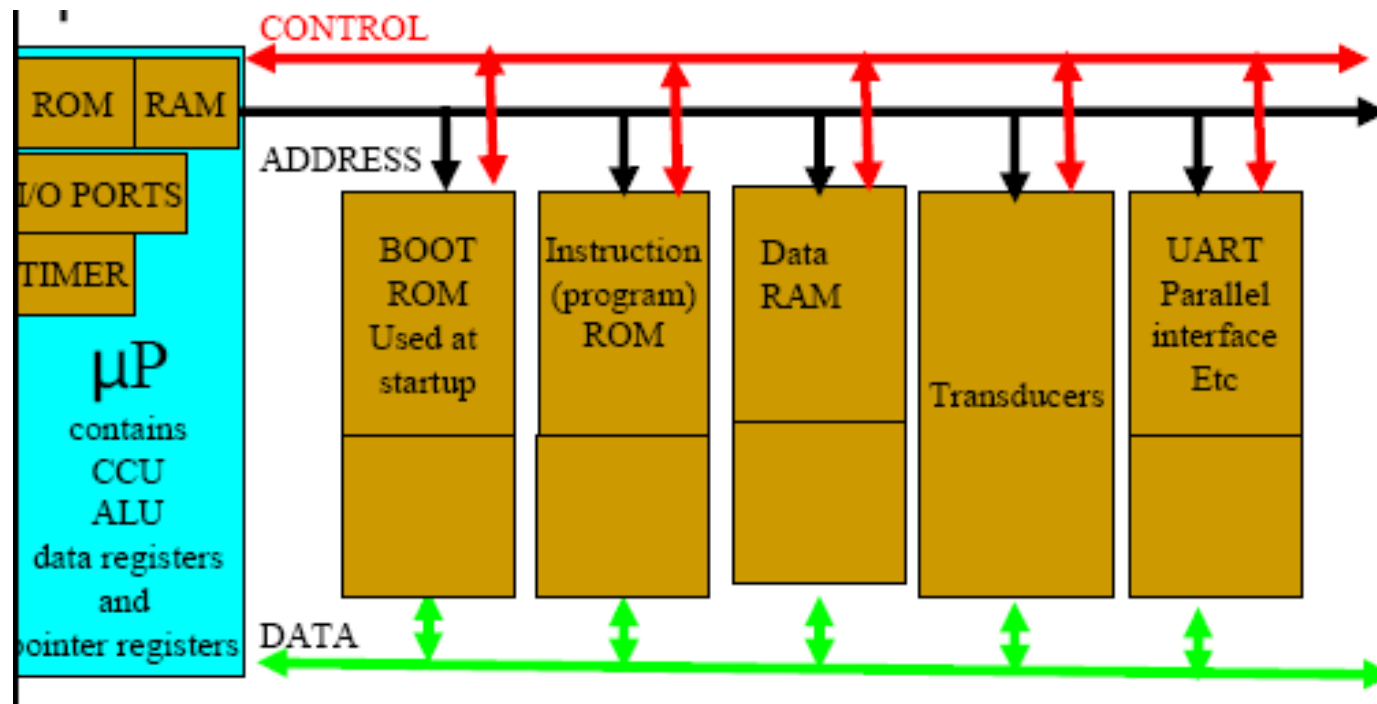
Many chips on mother board

Issues and Challenges for MP

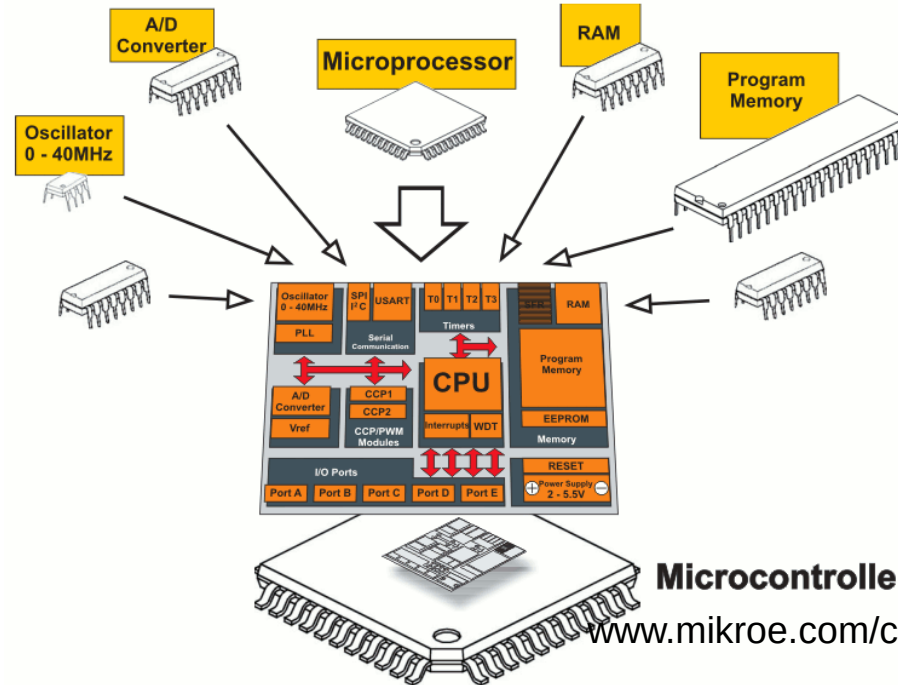
- ▶ Issues with increasing number of external devices or chips.
 - ▶ Many chips mean **many interconnections**
 - ▶ **Many interconnection** means -
 - ▶ Mechanical failure rates **increased**
 - ▶ Design time **increased**
 - ▶ Cost **increased**
 - ▶ Board size **increased**
 - ▶ Compatibility between parts **increased**

So ... Solution is Microcontroller !!

- ▶ Microcontroller – put a **limited** amount of most commonly used resources “inside” the chip
- ▶ A “**limited**” amount is often “**enough**” for many applications



What is a Microcontroller



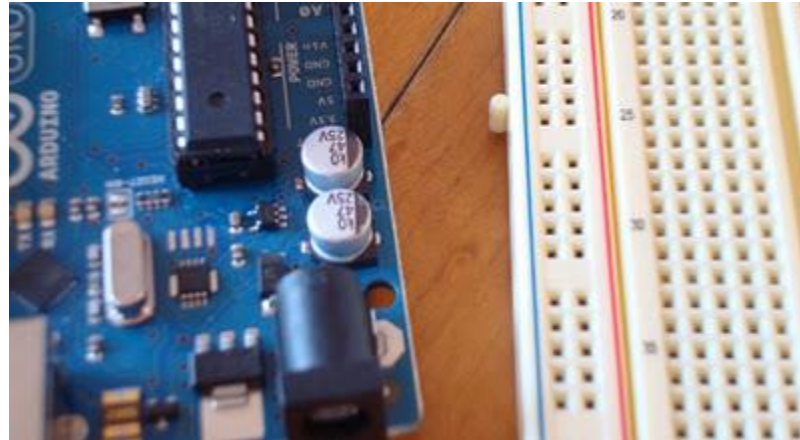
www.mikroe.com/chapters/view/1

Fig. 0-1 Microcontroller versus Microprocessor

- A small computer on a single chip
 - containing a processor, memory, and input/output
- Typically "**embedded**" inside some device that they control
- A microcontroller is often small and low cost

What is a Development Board

- ▶ A printed circuit board designed to facilitate work with a particular microcontroller.
- Typical components include:
 - power circuit
 - programming interface
 - basic input; usually buttons and LEDs
 - I/O pins



Advantages of MC over MP

- ▶ Number of Pins count **decreased**
- ▶ Design time **decreased**
- ▶ Board layout size **decreased**
- ▶ Upgrade path **easier** – matching between peripherals for speed
- ▶ Cost **decreased** – bulk purchases
- ▶ Reliability **increased**
- ▶ Common software / hardware design environment **available** from manufacturer

Microprocessor Vs Microcontroller

Microprocessor

- CPU is stand-alone, RAM, ROM, I/O, timer are separate
- designer can decide on the amount of ROM, RAM and I/O ports.
- expensive
- versatility
- general-purpose

Microcontroller

- CPU, RAM, ROM, I/O and timer are all on a single chip
- fix amount of on-chip ROM, RAM, I/O ports
- cost effective
- for applications in which cost, power and space are critical
- single-purpose

List of Microcontrollers

- ▶ **1972** - Texas Instrument TMS 1000, 1st single μ C, 4-bit
- ▶ **1976** - Intel 8048, 8-bit μ C, 1k ROM, 64b RAM, 27 I/O
- ▶ **1980** - Intel 8051, 4k ROM, 128b RAM, 32 I/O, 2 16-bits timers
- ▶ **1980s**
- ▶ (MCS-51 family)
 - ▶ Intel 8031, 8052, 8751, ...
 - ▶ Atmel AT89C51, AT 89C1052/2051, **Atmel AVR**
 - ▶ Dallas Semiconductor DS5000 series
 - ▶ Philips, National Semiconductor, ...
 - ▶ (Other μ Cs) Microchip PIC16 series, Motorola 68HC11, Zilog's Z86

AVR Microcontrollers

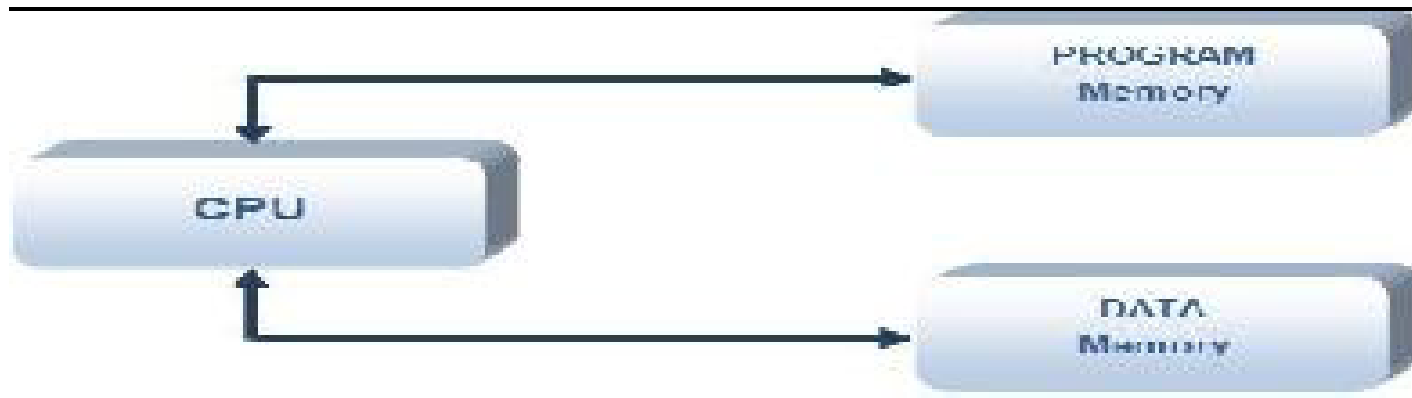
- ▶ AVR (**A**lf Egil Bogen and **V**egard Wollan's RISC processor) or AVR (**A**dvanced **V**irtual **R**ISC) microcontrollers are available in **three** categories:
- ▶ **Tiny AVR –**
 - ▶ Less memory, small size, suitable only for simpler applications.
- ▶ **Mega AVR –**
 - ▶ These are the most popular ones having good amount of memory (up to 256 KB), higher number of inbuilt peripherals and suitable for moderate to complex applications.
- ▶ **Xmega AVR –**
 - ▶ Used commercially for complex applications, which require large program memory and high speed.

AVR Microcontrollers

Series Name	# of Pins	Flash memory	Special Features
TinyAVR	6-32	0.5-8KB	Small in size
MegaAVR	28-100	4-256KB	Extended Peripherals
XmegaAVR	44-100	16-384KB	DMA, Event system included

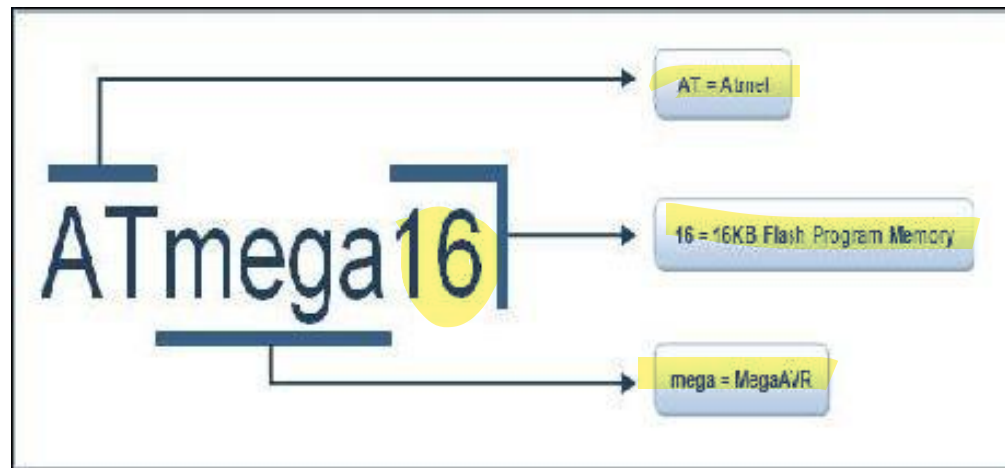
AVR Microcontrollers Features

- ▶ The **AVR --- a 8-bit RISC single chip** microcontroller developed by **Atmel** in 1996 --- uses modified Harvard architecture .
- ▶ In **Harvard architecture, program and data are stored in separate physical memory systems** that appear in different address spaces.



AVR Microcontrollers Features

- ▶ The AVR was one of the first microcontroller families to use on-chip flash memory for program storage.
- ▶ Whereas one-time programmable ROM, EPROM, or EEPROM used by other microcontrollers at the time.

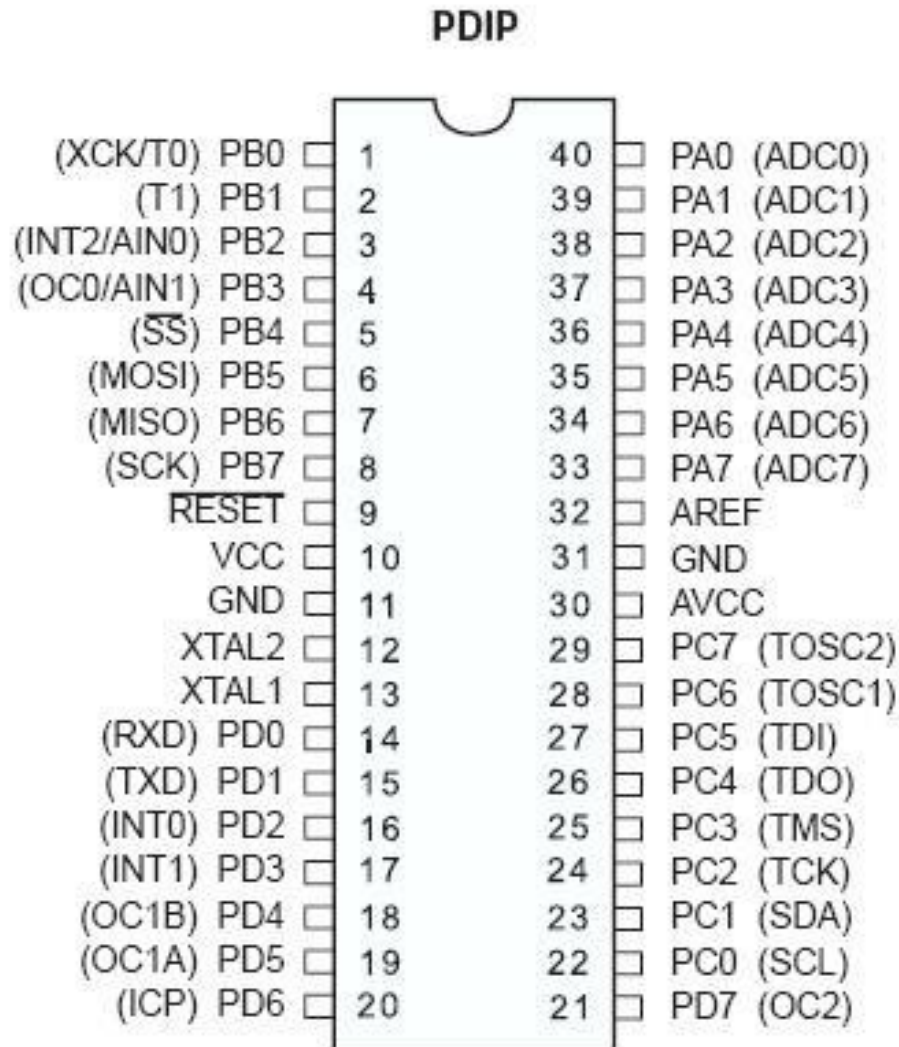


AVR Microcontroller Naming Convention

ATMega16: Features

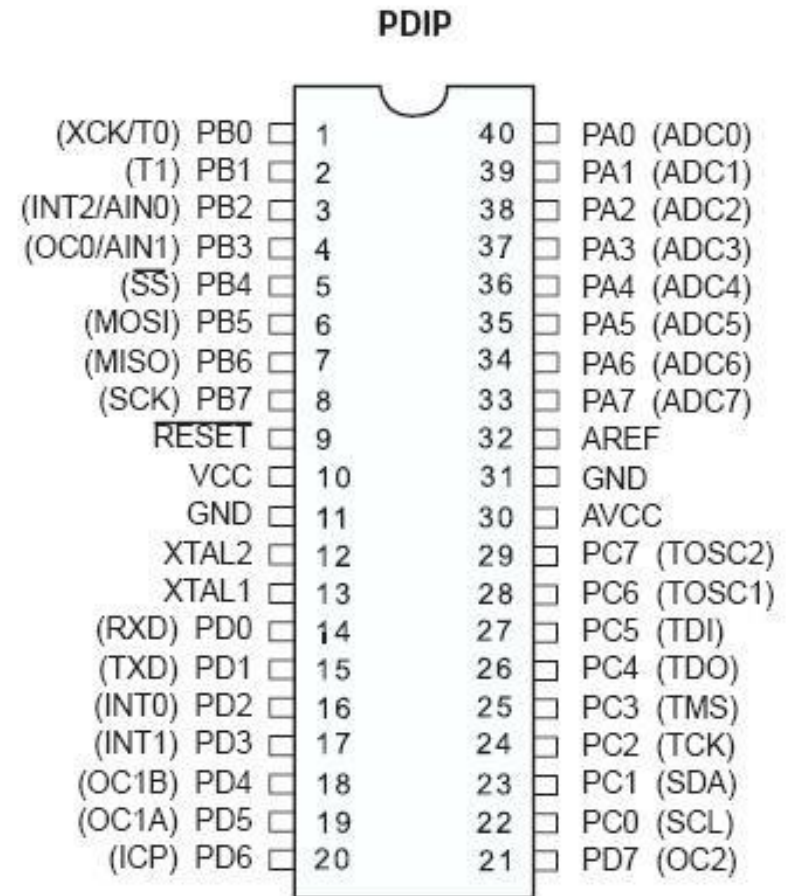
- ▶ 16 Kbytes of in-System Self-programmable Flash program memory
- ▶ 512 Bytes EEPROM
- ▶ 1 Kbyte Internal SRAM
- ▶ 32 8-bit General Purpose Working Registers
- ▶ 32 Programmable I/O Lines out of total 40 Pins DIP
- ▶ 8-channel, 10-bit ADC
- ▶ Two 8-bit Timer/Counters
- ▶ One 16-bit Timer/Counter
- ▶ Programmable Serial USART (Universal Synchronous Asynchronous Receiver Transmitter)
- ▶ 4 PWD (Pulse Width Modulator) Channel
- ▶ Operating Voltages: 4.5V - 5.5V
- ▶ Speed Grades: 0 - 16 MHz

ATMega16: Pin Descriptions



ATMega16: Pin Descriptions

- ▶ **GND:** Ground (0V). Note there are 2 ground Pins.
- ▶ **VCC:** Digital supply voltage. (+5V)
- ▶ **AVCC:** AVCC is the supply voltage pin for Port A and the A/D Converter.
- ▶ **AREF:** AREF is the analog reference pin for the A/D Converter.



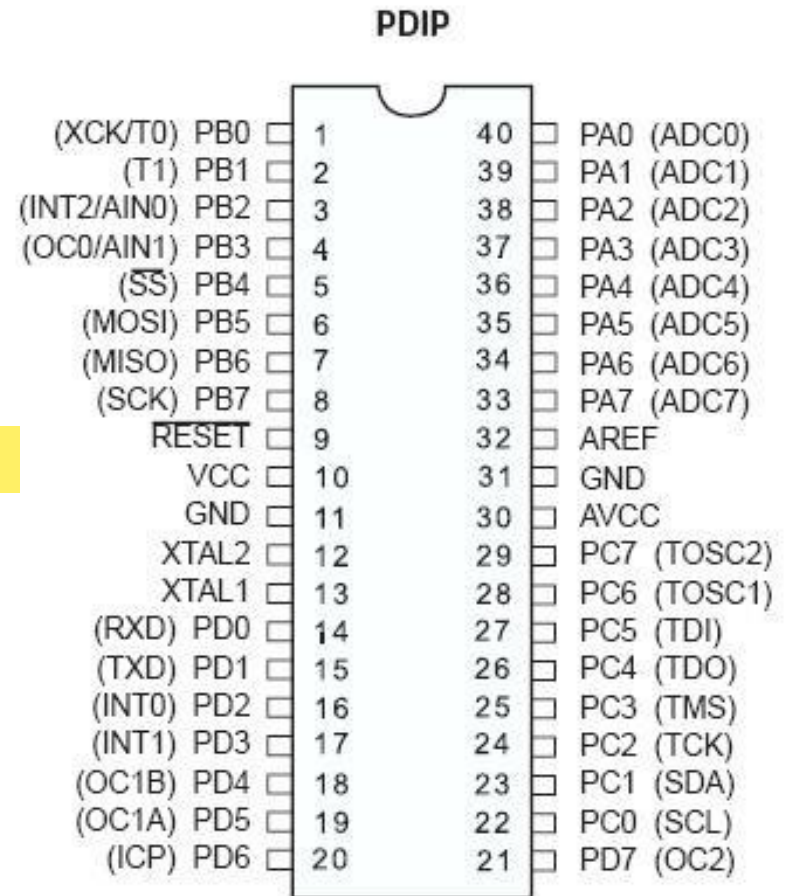
ATMega16: Pin Descriptions

▶ **XTAL1**: External oscillator pin 1

▶ **XTAL2**: External oscillator pin 2

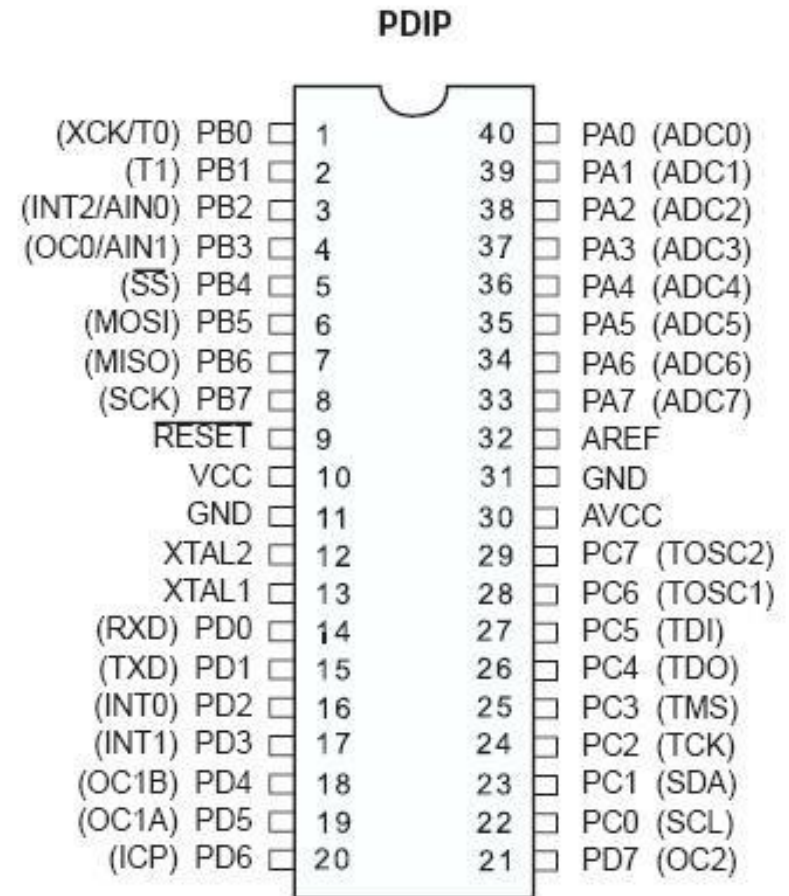
▶ **Port A (PA0 – PA7)**: Port A serves as the analog inputs to the A/D Converter.

▶ It also serves as an 8-bit bi-directional I/O port, if the A/D Converter is not used.

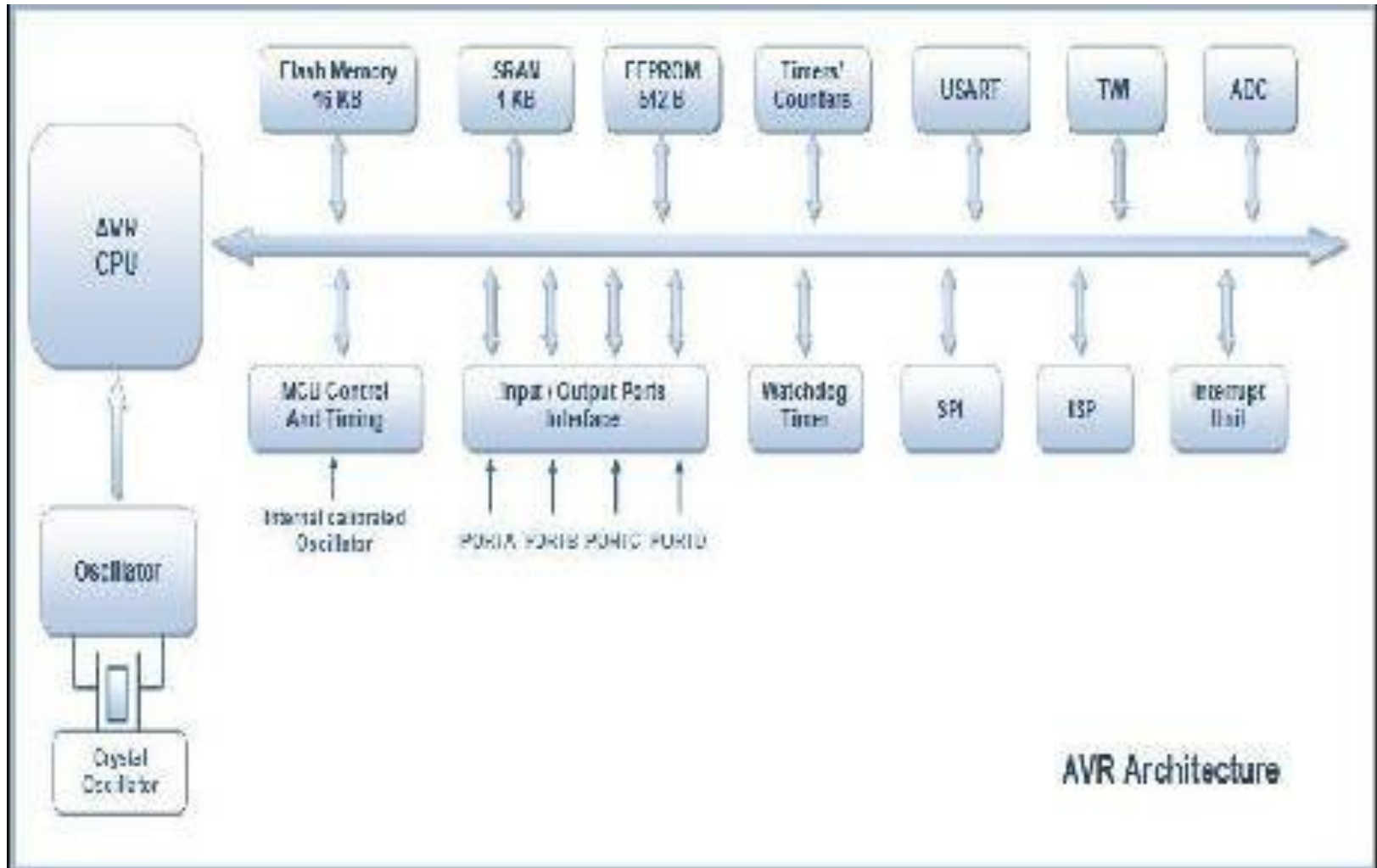


ATMega16: Pin Descriptions

- ▶ **Port B (PB0 – PB7):** Port B is an 8-bit bi-directional I/O port.
- ▶ **Port C (PB0 – PB7):** Port B is an 8-bit bi-directional I/O port.
- ▶ **Port D (PD0 – PD7):** Port B is an 8-bit bi-directional I/O port.
- ▶ Port B, C and D also serve the functions of various special features of the ATMega16.



ATMega16: Architectural Diagram



ATMega16: Architectural Diagram

► Oscillator:

- The ATMega16 can use an internal or external clock signal.
- By default Atmega16 is set to operate at internal calibrated oscillator of 1 MHz.
- The internal clock is an RC oscillator programmable to 1, 2, 4, or 8 MHz
- The maximum frequency of internal oscillator is 8Mhz.
- Alternatively, ATmega16 can be operated using an external crystal oscillator with a maximum frequency of 16MHz.

ATMega16: Architectural Diagram

▶ CPU:

- ▶ PC: Program Counter
 - ▶ Address of next instruction
- ▶ IR: Instruction Register
 - ▶ Pre-fetched instruction
- ▶ ID: Instruction decoder
 - ▶ Current instruction
- ▶ General Purpose Register (GPR): R0-R31
 - ▶ 8-bit x 32 GPR pipeline operation possible.
- ▶ ALU (Arithmetic Logic Unit)
- ▶ Status Register

ATMega16: Architectural Diagram

▶ CPU – Status Register:

Bit	7	6	5	4	3	2	1	0
	I	T	H	S	V	N	Z	C
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

▶ Bit 0 - C: Carry Flag

- ▶ The Carry Flag C indicates a carry in an arithmetic or logic operation.

▶ Bit 1 - Z: Zero Flag

- ▶ The Zero Flag Z indicates a zero result in an arithmetic or logic operation.

ATMega16: Architectural Diagram

▶ CPU – Status Register:

Bit	7	6	5	4	3	2	1	0
	I	T	H	S	V	N	Z	C
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

▶ Bit 2 - N: Negative Flag

- ▶ The Negative Flag N indicates a negative result in an arithmetic or logic operation.

▶ Bit 3 - V: Two's Complement Overflow Flag

- ▶ The Two's Complement Overflow Flag V supports two's complement arithmetic

ATMega16: Architectural Diagram

▶ CPU – Status Register:

Bit	7	6	5	4	3	2	1	0
	I	T	H	S	V	N	Z	C
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

▶ Bit 4 - S: Sign Bit Flag ($S = N \oplus V$)

- ▶ The S bit is always an XOR between the N and V value.

▶ Bit 5 - H: Half-carry Flag

- ▶ The Half Carry Flag H indicates a Half Carry in some arithmetic operations. Half Carry is useful in BCD arithmetic.

ATMega16: Architectural Diagram

▶ CPU – Status Register:

Bit	7	6	5	4	3	2	1	0
	I	T	H	S	V	N	Z	C
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

▶ Bit 6 - T: Bit Copy Storage Flag

- ▶ The Bit Copy instructions BLD (Bit Load) and BST (Bit Store) use the T-bit as source or destination for the operated bit.

▶ Bit 7 - I: Global Interrupt Enable Flag

- ▶ The Global Interrupt Enable bit must be set for the interrupts to be enabled.

ATMega16: Architectural Diagram

▶ **Timers/Counters:**

- ▶ Most commonly used complex peripheral
- ▶ Atmega16 consists of **two 8-bit and one 16-bit timer/counter.**
- ▶ Think of them as **binary up-counters**
 - ▶ In **timing mode, count time periods**
 - ▶ In **counting mode, counting events or pulses**
- ▶ Timers are useful for generating precision actions e.g., for creating time delays between two operations.

ATMega16: Architectural Diagram

▶ **USART:**

- ▶ **Universal Synchronous Asynchronous Receiver and Transmitter** interface is available for interfacing with external device capable of communicating serially (data transmission bit by bit).

▶ **Two Wire Interface (TWI):**

- ▶ TWI can be used to set up a network of devices
 - ▶ Many devices can be connected over TWI interface forming a network
 - ▶ The devices can simultaneously transmit and receive and have their own unique address.

ATMega16: Architectural Diagram

▶ **ADC Interface:**

- ▶ Atmega16 is equipped with an 8 channel ADC (**Analog to Digital Converter**) with a resolution of 10-bits.
- ▶ ADC reads the analog input for e.g., a sensor input and converts it into digital information which is understandable by the microcontroller.

▶ **Serial Peripheral Interface (SPI):**

- ▶ SPI port is used for serial communication between two devices on a common clock source.
- ▶ The data transmission rate of SPI is more than that of USART.

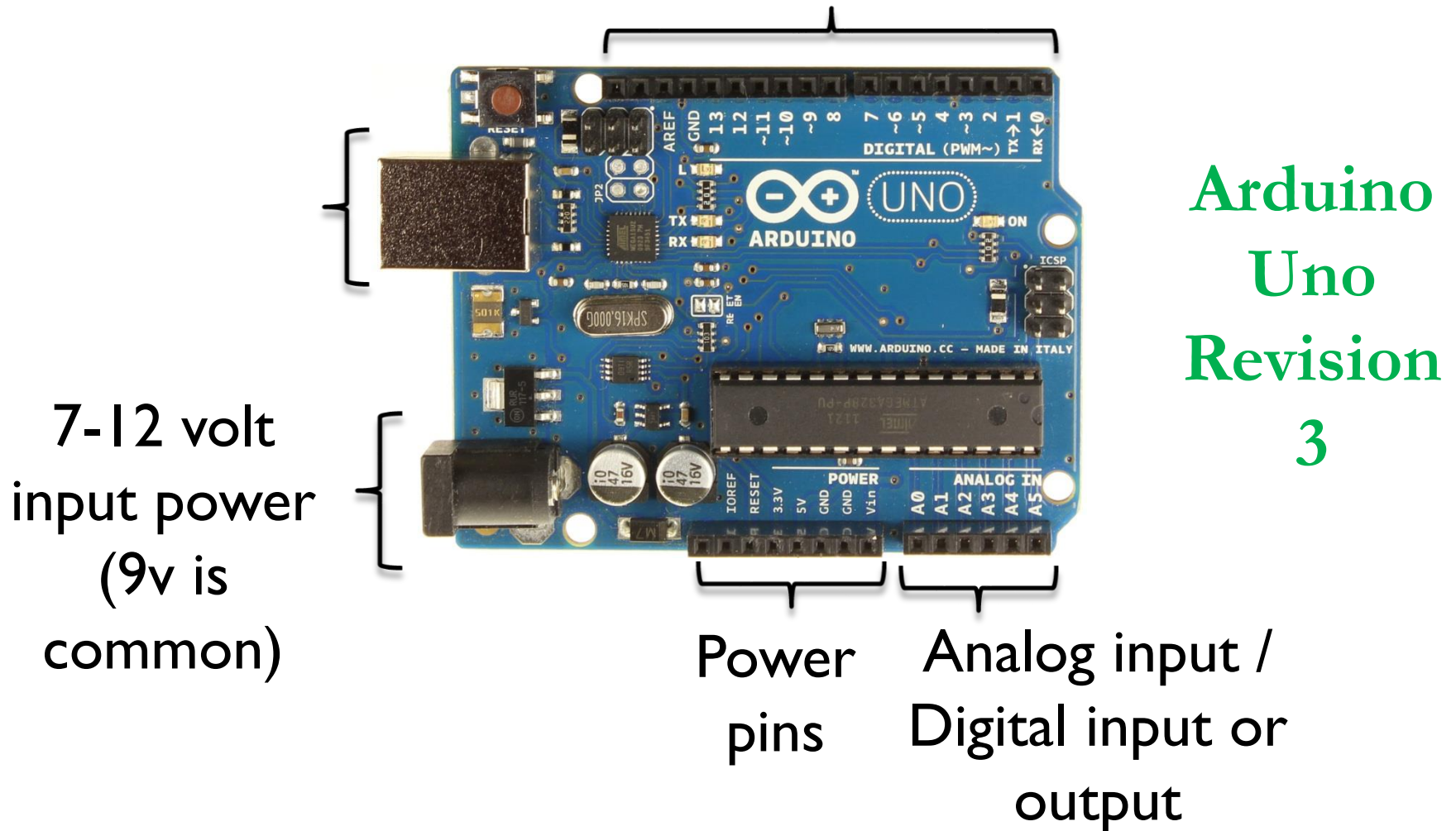
Why Was Arduino Developed?

- ▶ Physical Computing – using components that can interact with people and with the world around us
- ▶ The Arduino was originally developed for artists and designers to prototype interactive displays
 - ▶ Developed for non-scientists
 - ▶ Minimalist programming
 - ▶ “Forgiving” circuitry that can handle a wide variety of wiring errors

Different Varieties of Arduino



Digital Input / Digital output (PWM on pins 3, 5, 6, 9, 10, 11)



What Can Arduino Uno Be Used To Teach?

- ▶ Introductory electronics (voltage, current, resistance)
- ▶ How sensors and actuators work
- ▶ Simple programming
- ▶ Design of basic scientific equipment
- ▶ Troubleshooting
- ▶ Challenges of communicating with users through a project (e.g., messages, formatting numbers, ease of use, etc.)
- ▶ Statistics and variation in data gathering

What Can Students Do?

- ▶ Quickly able to prototype a working project
- ▶ Able to produce a working computer program
- ▶ The tactile “feel” of assembling a project is very rewarding; making something that works
- ▶ FINALLY understanding that a scientific instrument is only as good as its design and calibration
- ▶ Get to troubleshoot circuits under the tutelage of the instructor to develop a logical, orderly method.

What Are Some Good Components to Start With?

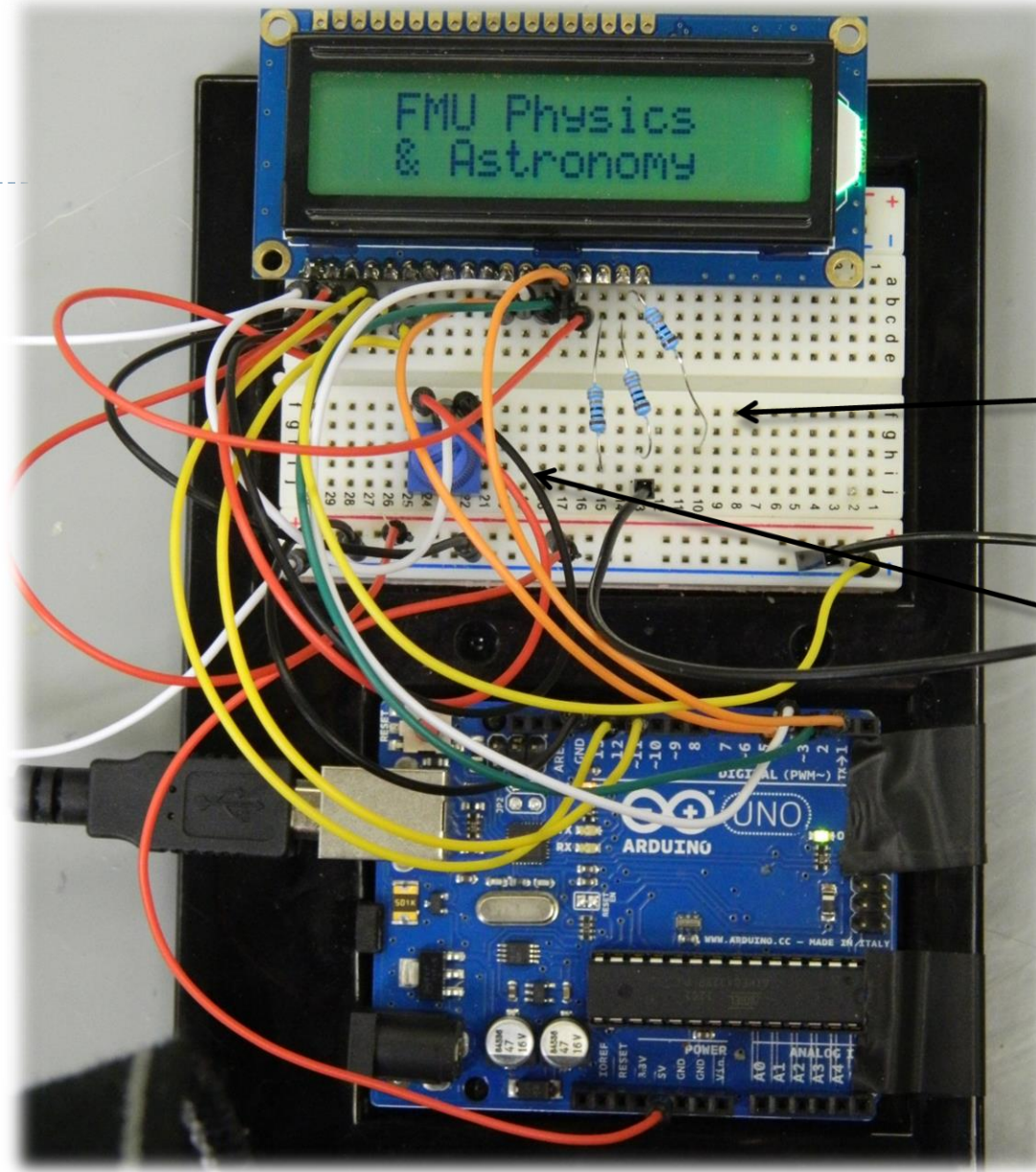
- ▶ LEDs....lots of LEDs with variety of colors (be sure to also acquire lots of current-limiting resistors)
- ▶ Good for blinking, dimming using PWM, “Knight Rider” effect with 8 or more LEDs, strobe effects, etc.



What Are Some Good Components to Start With?

- ▶ LCD Screen with backlight (16x2 is common and teaches the complications of textual display)
- ▶ The parallel version uses 4 digital pins for display
- ▶ Instructions for how to connect is built in as a sample “sketch” in the Arduino
- ▶ Students get a huge degree of satisfaction from transmitting text to the LCD screen



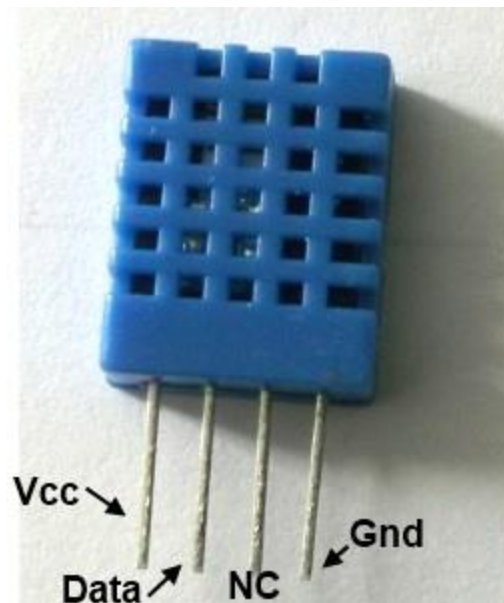


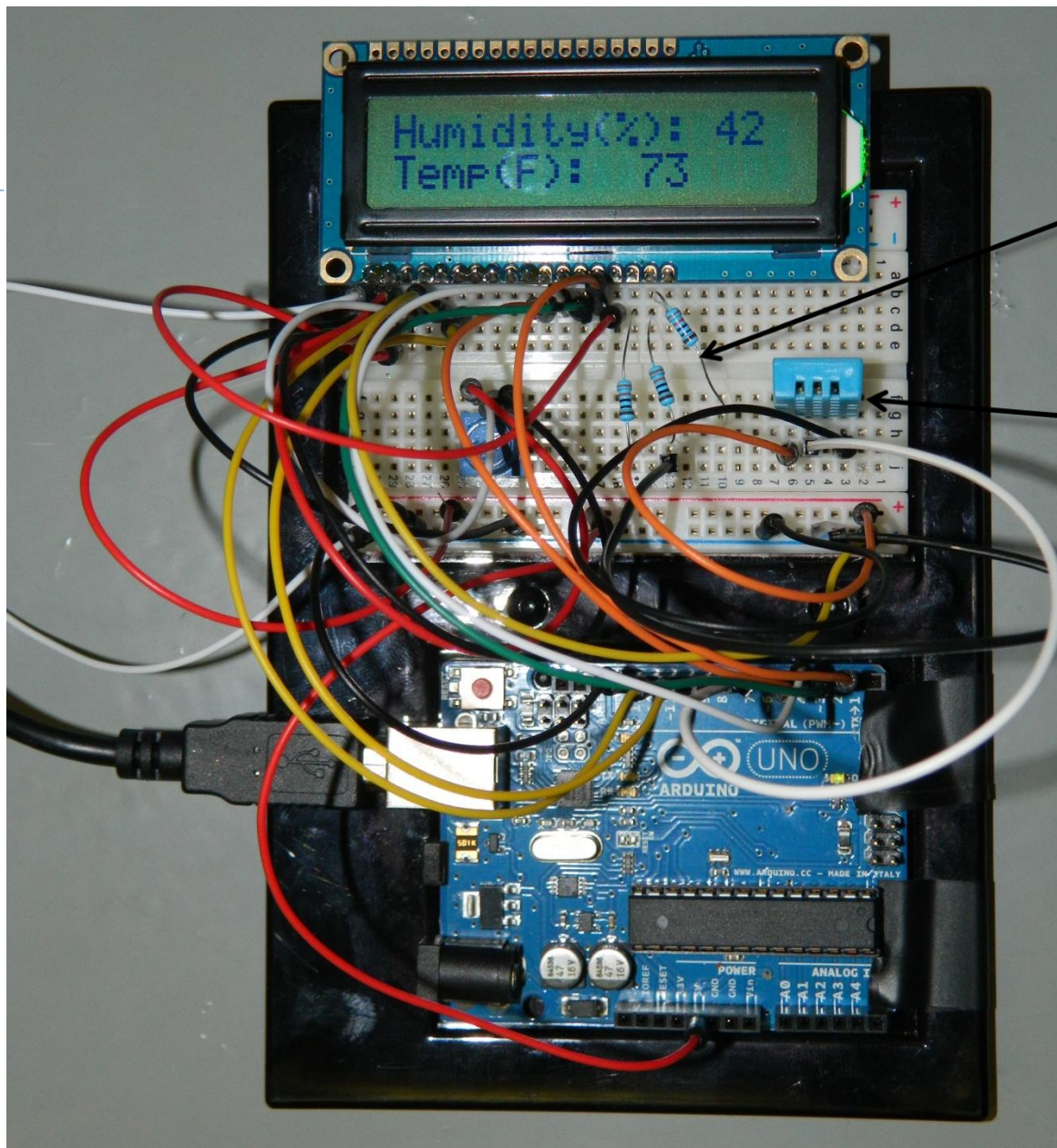
Current-limiting
resistors for RGB
backlight

10 k Ω
potentiometer to
adjust screen
contrast

What Are Some Good Components to Start With?

- ▶ Basic temperature and/or humidity sensor
- ▶ DHT11 or DHT22 is a good entry-level choice
- ▶ Library can be downloaded and incorporated to give easy access to features of the sensor





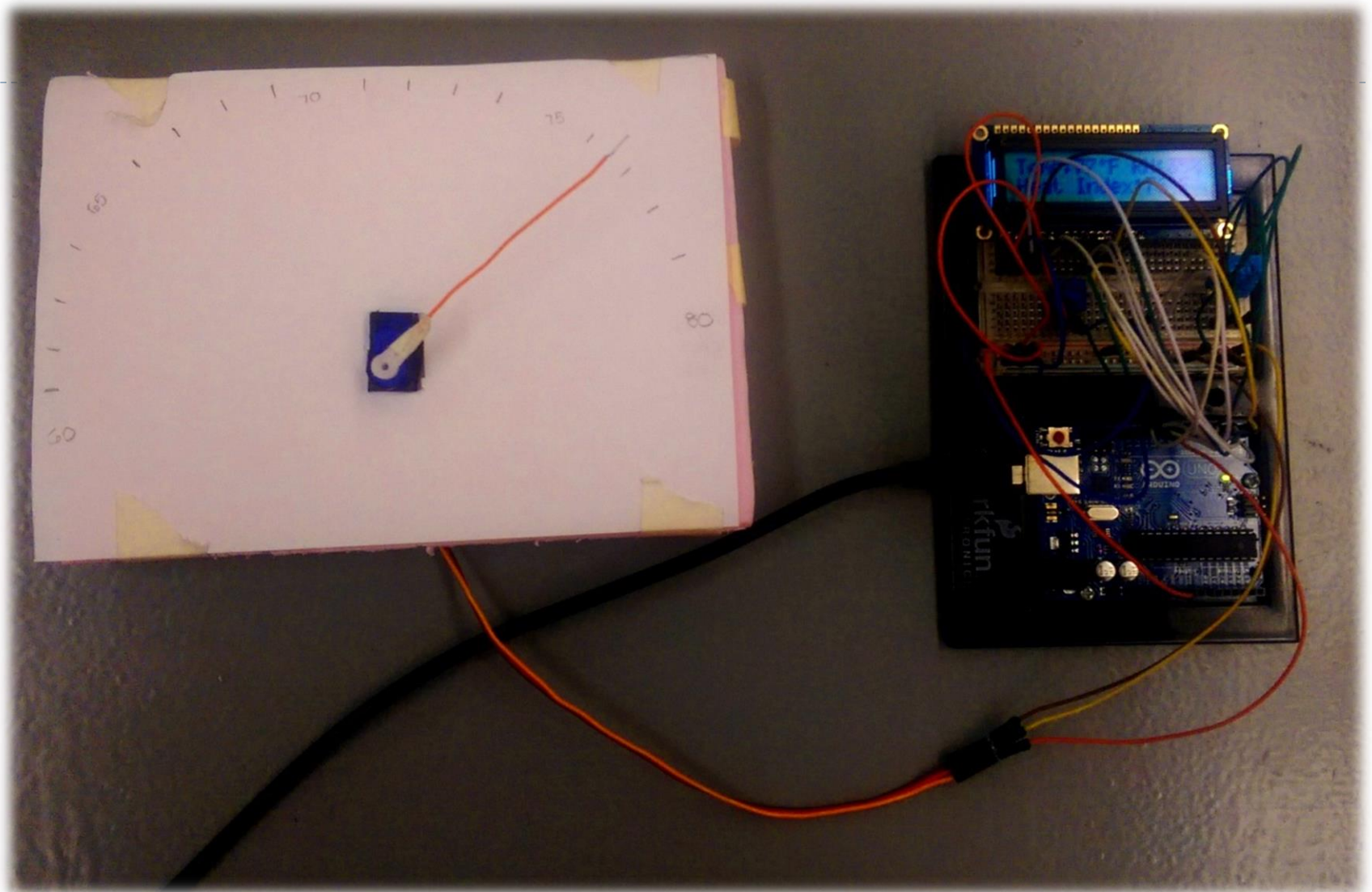
Current-limiting
resistors for RGB
backlight

DHT 11 Sensor

What Are Some Good Components to Start With?

- ▶ Small servos
- ▶ Arduino can be used to position servo at a given angle for use in projects.
- ▶ Other sensors can be attached to the servo to add a layer of complexity
- ▶ Servo library included in Arduino IDE

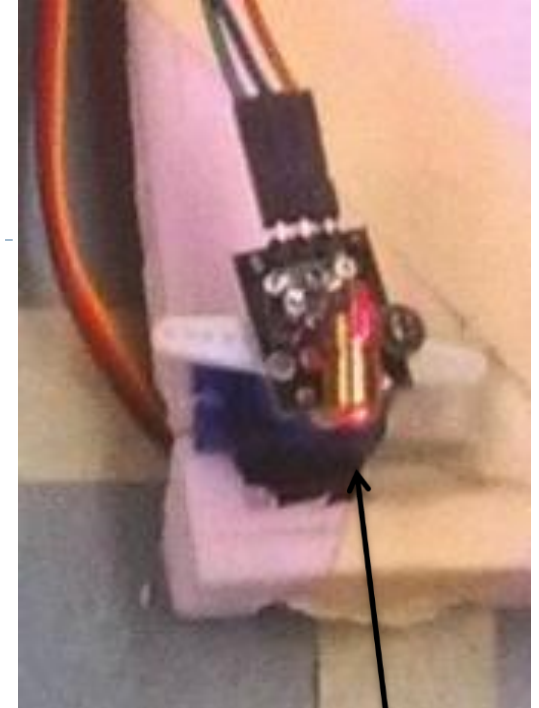
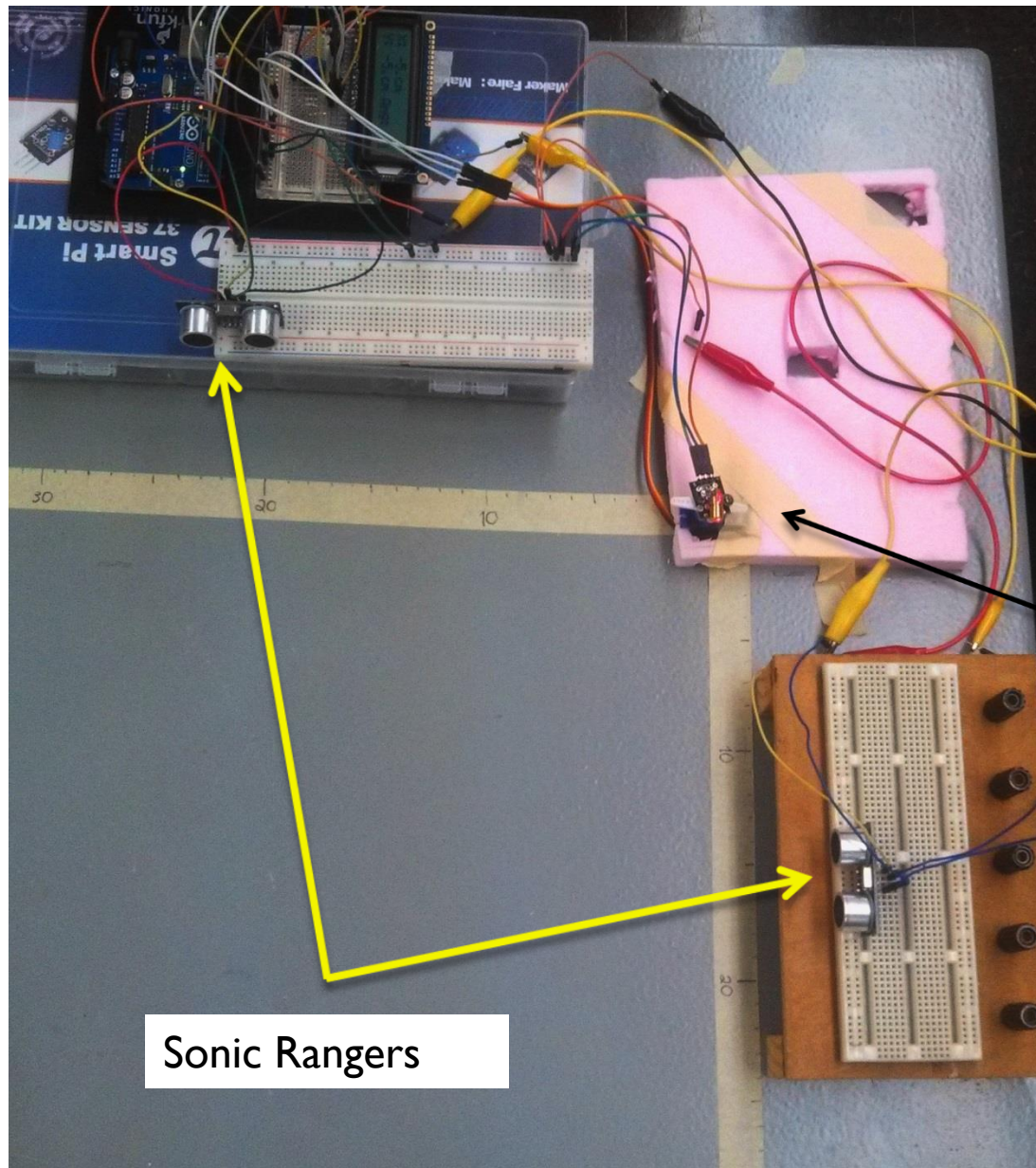




What Are Some Good Components to Start With?

- ▶ Ultrasonic Ranger – uses ultrasonic waves to determine the distance between sensor and object based on time between emission of wave and echo of it.
- ▶ Good for introducing discussion of speed of sound variation with environmental conditions
- ▶ Calibration principles are easily discussed



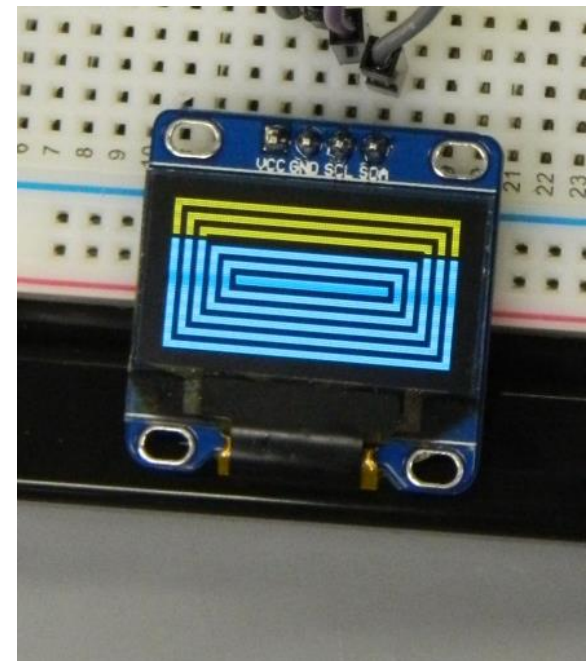
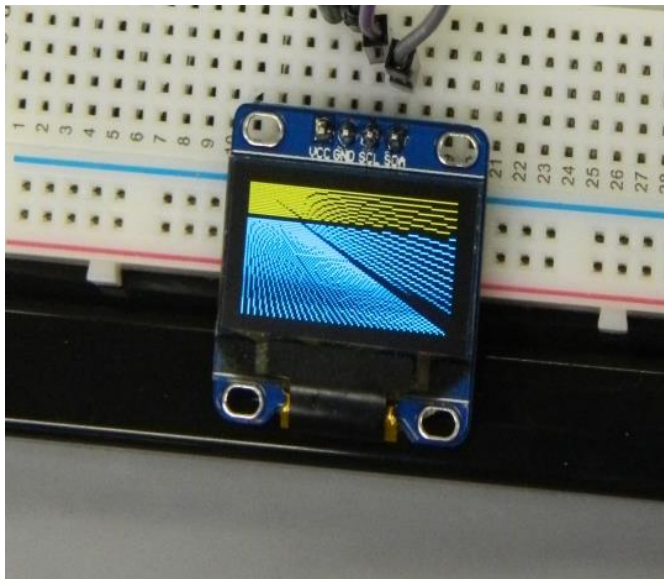


Sonic Rangers

Laser pointer mounted on servo that aims at a target based on readings of dual sonic rangers

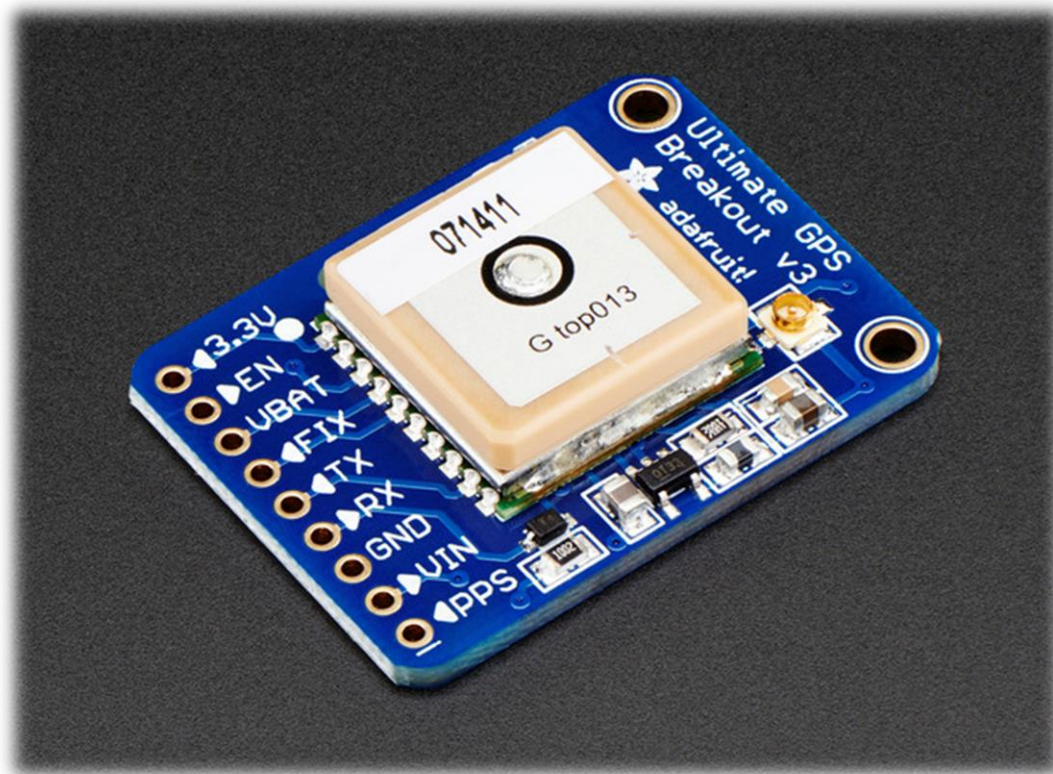
What Are Some Other Components That Are Relatively Easy to Use?

- ▶ 128x64 Organic LED screen that can handle graphics as well as text



What Are Some Other Components That Are Relatively Easy to Use?

- ▶ GPS Module that can be used for a variety of purposes such as triggering a response only when the user is standing in a particular location

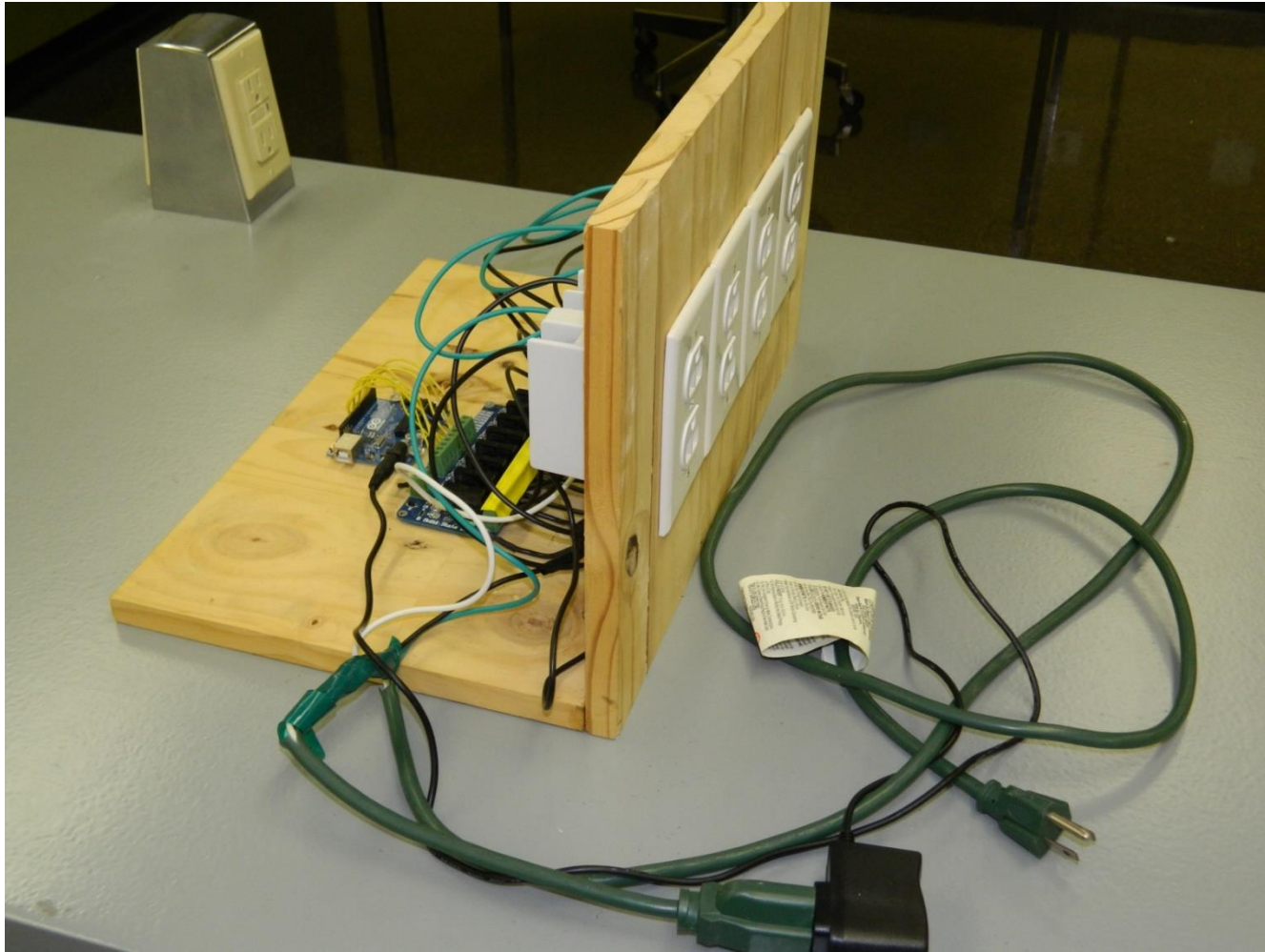


What Are Some Other Components That Are Relatively Easy to Use?

- * Mechanical or Solid-State Relay Boards to control alternating current loads of several amps



AC Outlets Controlled by Arduino



Thank You !!

